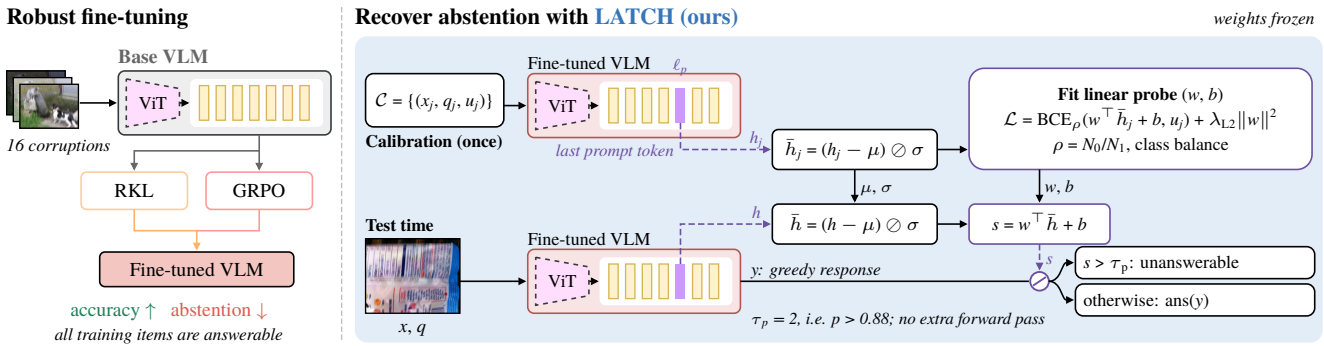

Robust fine-tuning
Base VLM
ViT
16 corruptions
RKL
GRPO
Fine-tuned VLM
accuracy ↑ abstention ↓
all training items are answerable
Recover abstention with LATCH (ours)
weights frozen
C = {(x_j, q_j, u_j)}
Calibration (once)
Fine-tuned VLM
ViT
ℓ_p
last prompt token
h_j
h̄_j = (h_j − μ) ⊘ σ
μ, σ
Fit linear probe (w, b)
L = BCE_ρ(w^⊤ h̄_j + b, u_j) + λ_L2 ||w||^2
ρ = N_0/N_1, class balance
w, b
Test time
h
h̄ = (h − μ) ⊘ σ
s = w^⊤ h̄ + b
s
x, q
y: greedy response
s > τ_p: unanswerable
otherwise: ans(y)
τ_p = 2, i.e. p > 0.88; no extra forward pass